

Notes: Easley, Kiefer, O’Hara, and Paperman (JF, 1996)

Liquidity, Information, and Infrequently Traded Stocks

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	Contribution relative to classic market microstructure theory	2
2.2	Contribution relative to spread decomposition and liquidity studies	2
2.3	Contribution relative to research on infrequently traded stocks	3
2.4	Contribution relative to later empirical microstructure	3
3	Model and economic mechanism	3
3.1	Trading environment	3
3.2	Figure 1: trading-process tree	4
3.3	Market maker pricing	4
3.4	Probability of information-based trading	5
3.5	Spread implications	5
4	Data and sample construction	5
4.1	NYSE stock universe	5
4.2	Price-matched sample design	5
4.3	Trade classification	6
5	Estimation strategy	6
5.1	Likelihood function	6
5.2	Parameter restrictions	6
6	Identification and interpretation	7
6.1	What identifies information risk	7
6.2	Why spread regressions validate the model	7
6.3	Limitations	7
7	Tables and figures	8
7.1	Figure 1 and Table I: model structure and estimated parameters	8
7.2	Table II and Figure 2: nonparametric tests and PIN distributions	8
7.3	Table III and Figure 3: spread behavior	9
7.4	Table IV: regression results	10
7.5	Appendix tables: stock-level estimates and sample composition	10
8	Short summary	12
9	Conclusion	12
9.1	Core conclusion	12
9.2	Economic intuition	12
9.3	Takeaway	12

1 Big picture

- **Question.** Do differences in asymmetric-information risk help explain why infrequently traded stocks have larger bid-ask spreads than actively traded stocks?
- **Core idea.** Even if a stock trades rarely, the trades that do occur may be disproportionately information based. A market maker who faces a higher probability of trading with informed investors sets wider spreads.
- **Empirical object.** The paper estimates the probability that a randomly arriving trade is information based, now commonly called **PIN**, using only the daily counts of buyer-initiated and seller-initiated trades.
- **Sample.** The authors study 90 NYSE stocks from October 1 to December 23, 1990: 30 stocks from the highest volume decile, 30 from the fifth decile, and 30 from the eighth decile.
- **Matching design.** The sample is price matched across volume deciles so that differences in spreads are not mechanically driven by different stock price levels.
- **Main result.** The probability of information-based trading is lower for the most active stocks than for less active stocks. However, the fifth and eighth deciles have very similar information-based trading risk.
- **Spread result.** Spreads are lower for the most active stocks. They are not statistically different between the fifth and eighth deciles. This pattern matches the estimated PIN pattern.
- **Regression result.** A regression of quoted spread on $V \cdot \text{PIN}$, where V is average price, and volume shows that $V \cdot \text{PIN}$ strongly predicts spreads. Volume alone explains much less.
- **Policy implication.** The paper argues that low-volume stocks are not simply costly because they are illiquid. They may be costly because a larger fraction of their trades is information based.
- **Takeaway.** Liquidity, trading volume, and asymmetric information are tightly linked. Market makers do not just charge for inventory risk or fixed operating costs; they also price the expected losses from informed trading.

2 Incremental contribution of the paper

2.1 Contribution relative to classic market microstructure theory

- Before this paper, theoretical models such as Glosten–Milgrom, Kyle, and Easley–O’Hara showed that asymmetric information creates bid-ask spreads.
- Those models made clear qualitative predictions, but empirical researchers had limited tools for estimating the probability of informed trading directly from transaction data.
- Easley, Kiefer, O’Hara, and Paperman provide a structural likelihood framework that maps observable buy and sell counts into estimates of private-information arrival rates.

Interpretation: The incremental step is to transform asymmetric-information theory into an estimable empirical object. The paper provides one of the early empirical routes from trade counts to information risk.

2.2 Contribution relative to spread decomposition and liquidity studies

- Previous empirical work documented that less active stocks have larger spreads, but the interpretation was ambiguous.
- Larger spreads could reflect inventory risk, order-processing costs, monopoly power, tick size effects, or asymmetric information.

- This paper isolates the information component by estimating PIN and showing that the cross-decile spread pattern lines up with estimated information risk.

Interpretation: The paper’s contribution is not merely to document a spread-volume relation. It provides a mechanism: low-volume stocks face a higher probability that a given trade comes from informed investors.

2.3 Contribution relative to research on infrequently traded stocks

- Infrequently traded securities had often been viewed mainly as illiquid assets with inventory or search frictions.
- The paper shows that infrequent trading can coexist with a high informed-trade probability.
- It also shows that the least active stocks in the sample are not always more information risky than moderately traded stocks once price is controlled.

Interpretation: This contribution is nuanced: low volume and information risk are related, but the relation is not mechanically monotone across all low-volume categories.

2.4 Contribution relative to later empirical microstructure

- The paper introduces an empirical object that became widely used: the probability of information-based trading, or PIN.
- It demonstrates how to estimate latent information events, informed trade arrival rates, and uninformed trade arrival rates from observable trade direction data.
- It anticipates later work that uses order-flow imbalance and trade arrivals to measure adverse selection risk, information risk, and liquidity provision costs.

Interpretation: The paper is important because it made private information risk measurable at the stock level in a way that could be used in cross-sectional tests.

3 Model and economic mechanism

3.1 Trading environment

The trading day is described by a latent information-event tree. At the start of each day, nature determines whether there is a private information event.

- With probability α , an information event occurs.
- Conditional on an event, the signal is bad news with probability δ and good news with probability $1 - \delta$.
- If no information event occurs, only uninformed buyers and sellers arrive.
- Uninformed buy and sell orders arrive according to Poisson processes with rate ϵ .
- Informed traders arrive at rate μ on information-event days and trade in the direction of their information.

Interpretation: This model generates observable daily buy and sell counts from unobservable information events. The market maker does not observe whether a day has news, but order flow helps him update beliefs.

3.2 Figure 1: trading-process tree

Read:

- Nature first chooses whether an information event occurs.
- If an event occurs, the signal is either good or bad.
- On good-news days, informed traders arrive as buyers; on bad-news days, they arrive as sellers.
- On no-news days, only uninformed buyers and sellers arrive.

Economic message: The tree shows why order imbalance is informative. A day with many buys may indicate good news; a day with many sells may indicate bad news.

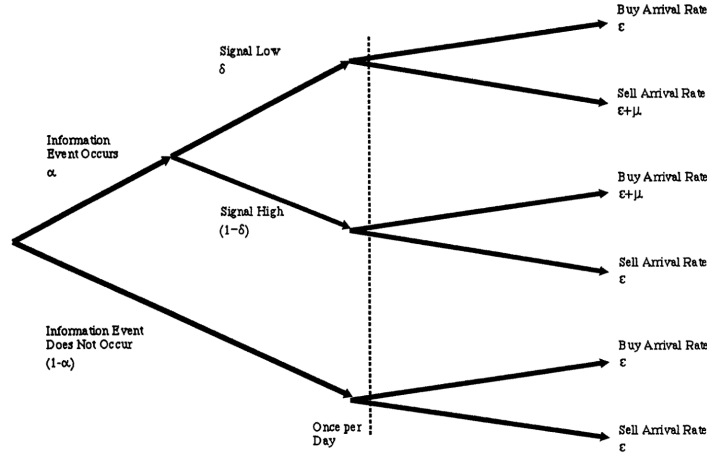


Figure 1. Tree diagram of the trading process. This figure gives the structure of the trading process, where α is the probability of an information event, δ is the probability of a low signal, μ is the rate of informed trade arrival, and ϵ is the rate of uninformed buy and sell trade arrivals. Nodes to the left of the dotted line occur once per day.

3.3 Market maker pricing

The market maker is competitive and risk neutral. He sets zero-expected-profit quotes conditional on the history of trade arrivals and the arriving order type.

Let the market maker's time- t belief vector be

$$P(t) = (P_n(t), P_b(t), P_g(t)), \quad (1)$$

where n denotes no news, b bad news, and g good news.

The expected value of the asset before observing the next trade is

$$E[V_i | t] = P_n(t)V_i^* + P_b(t)V_i^L + P_g(t)V_i^H. \quad (2)$$

The bid is below this expectation because a sell order increases the posterior probability of bad news. The ask is above this expectation because a buy order increases the posterior probability of good news.

Interpretation: Spreads compensate the market maker for the adverse-selection loss generated by informed traders.

3.4 Probability of information-based trading

At a given time t , the probability that a trade is information based depends on the probability of a news state and the relative arrival rates of informed and uninformed traders. The empirical stock-level version is usually summarized as

$$\text{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}. \quad (3)$$

- The numerator is the expected arrival rate of informed orders.
- The denominator is the expected arrival rate of all orders: informed plus uninformed buys and sells.
- Higher α or μ increases PIN.
- Higher uninformed trading ε decreases PIN.

Interpretation: A stock can be low volume because uninformed arrival rates are low. If informed arrivals do not fall proportionally, PIN rises.

3.5 Spread implications

The model implies that the spread depends positively on information-based trading risk. In simplified form, the opening spread is proportional to the probability that the first trade is informed multiplied by the value range of the asset:

$$\Sigma \approx (V_i^H - V_i^L) \text{PIN}_i. \quad (4)$$

For empirical testing, the paper uses the regression form

$$\Sigma_i = \beta_0 + \beta_1 V_i \cdot \text{PIN}_i + \beta_2 \text{Vol}_i + \eta_i, \quad (5)$$

where Σ_i is the quoted spread, V_i is stock price, and Vol_i is dollar trading volume.

Interpretation: If the model is economically meaningful, $V_i \cdot \text{PIN}_i$ should have a positive coefficient in spread regressions. This is exactly what the paper finds.

4 Data and sample construction

4.1 NYSE stock universe

- The paper starts from NYSE-listed common stocks.
- It excludes preferred stock, rights, warrants, stock funds, and ADRs.
- Stocks are ranked by total 1990 trading volume.
- The authors focus on the first, fifth, and eighth volume deciles.

Interpretation: The first decile represents actively traded stocks. The fifth and eighth deciles represent increasingly less active stocks while still having enough trades for estimation.

4.2 Price-matched sample design

The paper constructs a price-matched sample to avoid confounding price levels with trading volume.

- Average closing prices are computed from CRSP for October 1 to December 23, 1990.
- Stocks from the first and fifth volume deciles are matched by price.
- Thirty matched pairs are randomly selected.

- For each pair, the eighth-decile stock with the nearest price is selected.
- The final sample contains 90 stocks: 30 from each of the first, fifth, and eighth volume deciles.

Interpretation: Controlling for price is important because spreads can mechanically vary with the level of the stock price.

4.3 Trade classification

Trade data come from the ISSM database for October 1 to December 23, 1990. Trades are classified as buys and sells using a Lee–Ready-style procedure.

- Trades above the quote midpoint are classified as buys.
- Trades below the midpoint are classified as sells.
- Trades at the midpoint are classified using the tick test.
- Trades within five seconds at the same price without intervening quote revisions are collapsed to avoid multiple reports of one economic trade.

Interpretation: The structural likelihood is estimated from daily counts of buyer-initiated and seller-initiated trades.

5 Estimation strategy

5.1 Likelihood function

For each stock and each day, the likelihood integrates over the three latent states: no news, bad news, and good news. Let B_d and S_d be the number of buy and sell orders on day d . The day-level likelihood has the structure

$$L_d(\theta) = (1 - \alpha) \Pr(B_d, S_d \mid \text{no news}) \quad (6)$$

$$+ \alpha \delta \Pr(B_d, S_d \mid \text{bad news}) \quad (7)$$

$$+ \alpha(1 - \delta) \Pr(B_d, S_d \mid \text{good news}), \quad (8)$$

where $\theta = (\alpha, \delta, \varepsilon, \mu)$.

The sample likelihood is the product over days:

$$L(\theta) = \prod_d L_d(\theta). \quad (9)$$

Interpretation: Because the true information state is unobserved, the likelihood is a mixture of Poisson arrival processes.

5.2 Parameter restrictions

- The probabilities α and δ are restricted to $(0, 1)$ using transformations.
- The arrival rates ε and μ are restricted to be positive.
- Standard errors are obtained using the delta method.

Interpretation: The model is estimated separately for each stock, producing stock-specific information-risk estimates.

6 Identification and interpretation

6.1 What identifies information risk

- Symmetric high buy and sell counts are consistent with high uninformed trading.
- Days with unusually high buy counts are consistent with good-news events.
- Days with unusually high sell counts are consistent with bad-news events.
- The frequency of such imbalanced days helps identify α .
- The magnitude of order imbalance helps identify μ and δ .

Interpretation: Identification comes from the time-series pattern of buy and sell counts, especially order-flow imbalance.

6.2 Why spread regressions validate the model

The model uses trade data to estimate PIN, while the spread tests use quote and price data. This separation is important.

- If the structural estimates were meaningless, they would not predict independent spread data.
- The paper shows that $V \cdot \text{PIN}$ strongly predicts spreads.
- The regression evidence therefore validates the economic content of the PIN estimates.

Interpretation: The regression links an unobserved information-risk estimate to an observed market outcome.

6.3 Limitations

- Trade classification can misclassify some buys and sells.
- The model assumes independent days and stationary parameters over the estimation window.
- It abstracts from inventory costs, order-processing costs, and market power.
- The sample contains only 90 stocks and covers one short period in 1990.

Interpretation: The paper does not claim that information risk is the only determinant of spreads. It shows that information risk is economically important and helps explain cross-sectional spread differences.

7 Tables and figures

7.1 Figure 1 and Table I: model structure and estimated parameters

Read:

- Figure 1 displays the latent trading process used in the model.
- Table I reports mean, median, and standard deviation of estimated parameters by volume decile.
- The informed arrival rate μ is much higher in the first decile than in the fifth and eighth deciles.
- Uninformed arrival rate ε falls dramatically with volume, from 0.176 in the first decile to 0.0096 in the eighth decile.
- The average PIN is lower in the first decile than in the fifth and eighth deciles.

Economic message: Active stocks have many more uninformed trades. This lowers the probability that any given trade is informed.

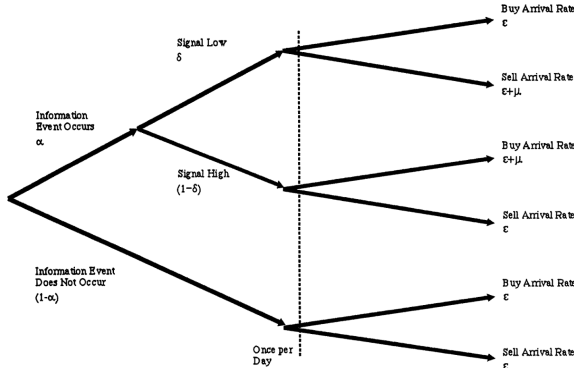


Figure 1. Tree diagram of the trading process. This figure gives the structure of the trading process, where α is the probability of an information event, δ is the probability of a low signal, μ is the rate of informed trade arrival, and ε is the rate of uninformed buy and sell trade arrivals. Nodes to the left of the dotted line occur once per day.

Figure 1: trading process tree

Table I
Summary Parameter Estimate Statistics by Decile

This table presents means, medians, and sample standard deviations of parameter estimates by volume decile for the 90 stocks in our sample. The parameter μ is the arrival rate of informed traders, ε is the arrival rate of uninformed traders, α is the probability of an information event, and δ is the probability that new information is bad news. The parameter Prob (Inf) is a composite variable measuring the probability of information-based trade.

Parameter	First Decile	Fifth Decile	Eighth Decile
Number in Sample	30	30	30
μ			
Mean	0.131970	0.030148	0.015696
Median	0.104864	0.027596	0.014122
Std. dev.	0.079314	0.013238	0.008607
ε			
Mean	0.175742	0.023970	0.009614
Median	0.136797	0.022917	0.008925
Std. dev.	0.141192	0.013158	0.005093
α			
Mean	0.500294	0.433952	0.356320
Median	0.477761	0.448613	0.363841
Std. dev.	0.141192	0.170253	0.173540
δ			
Mean	0.349078	0.444393	0.501787
Median	0.360357	0.418164	0.455418
Std. dev.	0.227188	0.238763	0.318183
Prob(Inf)			
Mean	0.163919	0.207788	0.220245
Median	0.154193	0.205858	0.196712
Std. dev.	0.043794	0.064794	0.121155

Table I: parameter estimates by decile

7.2 Table II and Figure 2: nonparametric tests and PIN distributions

Read:

- Table II reports Kruskal–Wallis and Mann–Whitney tests for differences across deciles.
- The arrival rates μ and ε differ strongly across all three deciles.
- Differences in PIN are significant between the first and fifth deciles, and between the first and eighth deciles.
- Differences in PIN between the fifth and eighth deciles are not significant.
- Figure 2 shows this visually: the fifth- and eighth-decile PIN distributions are similar, while the first-decile distribution lies to the left.

Economic message: The information-risk difference is between active stocks and less-active stocks, not between the moderately inactive and most inactive stocks in the sample.

Table II
Nonparametric Tests

The Kruskal-Wallis statistic is used to test the null hypothesis that parameter values for all three volume samples are drawn from identical populations versus the alternative hypothesis that at least one of the populations tends to furnish greater observed values than other populations. The parameter μ is the arrival rate of informed traders, ε is the arrival rate of uninformed traders, α is the probability of an information event, and δ is the probability that new information is bad news. The parameter Prob (Inf) is a composite variable measuring the probability of information-based trade.

Panel A: Kruskal-Wallis Tests on Parameters	
Parameter	Test Statistic
μ	66.279
ε	69.859
α	10.853
δ	4.236
Prob(Inf)	8.027

Critical value for $\alpha = 0.05$ is 5.991.

The Mann-Whitney statistic is used to test the null hypothesis that two samples are drawn from identical populations against the alternative that one population tends to yield higher values. The parameter μ is the arrival rate of informed traders, ε is the arrival rate of uninformed traders, α is the probability of an information event, and δ is the probability that new information is bad news. The parameter Prob (Inf) is a composite variable measuring the probability of information-based trade.

Panel B: Mann-Whitney Tests on Parameters			
Pairwise Comparisons (n = 30, m = 30)			
Parameter	1 to 5	1 to 8	5 to 8
μ	6.402	6.623	4.480
ε	6.505	6.653	5.071
α	1.390	3.326	1.789
δ	-1.508	-1.937	-0.547
Prob(Inf)	-2.883	-1.952	0.192

The test statistic is normally distributed and the critical value for $\alpha = 0.05$ is ± 1.6449 .

Table II: nonparametric tests

7.3 Table III and Figure 3: spread behavior

Read:

- Table III reports average quoted spreads and average percentage spreads by volume decile.
- The average spread rises from 0.176 in the first decile to 0.255 in the fifth and 0.271 in the eighth.
- Percentage spreads also rise from the first decile to the fifth and eighth deciles.
- Figure 3 plots percentage spread against stock price by volume decile.
- The first-decile curve lies below the fifth and eighth curves, while the fifth and eighth curves are similar.

Economic message: The spread pattern mirrors the information-risk pattern. The most active stocks have lower PIN and lower spreads; the fifth and eighth deciles have similar PIN and similar spreads.

Table III Spread Summary Statistics by Volume Decile			
Descriptive statistics are presented in this table for the 90 stocks included in our sample reported by 1990 volume decile. Average spread is the time weighted mean of the New York Stock Exchange (NYSE) quoted spread from the Institute for the Study of Security Markets (ISSM) database for the period of 10/1/90 to 12/23/90. Average percentage spread is the time-weighted mean of the NYSE quoted spread divided by the midpoint of the quote from the ISSM database for the period of 10/1/90 to 12/23/90.			
	First Decile	Fifth Decile	Eighth Decile
Number in sample	30	30	30
Average spread			
Mean	0.1763	0.2549	0.2708
Median	0.1717	0.2581	0.2802
Std. dev.	0.0243	0.0588	0.0585
Average % spread			
Mean	1.4140	1.9158	1.9824
Median	0.7123	1.1446	1.1688
Std. dev.	1.6961	1.9379	2.0211

Table III: spread summary statistics

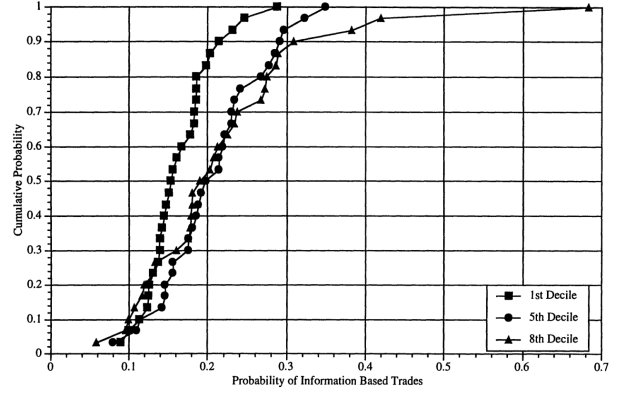


Figure 2: Cumulative distribution of the probability of information-based trades by volume decile. This graph shows the cumulative probability distribution of the probability that a trade comes from an informed trader, $P(\text{Inf}|\text{Trade})$, for each of the three 1990 volume deciles in our sample.

Figure 2: cumulative distribution of PIN

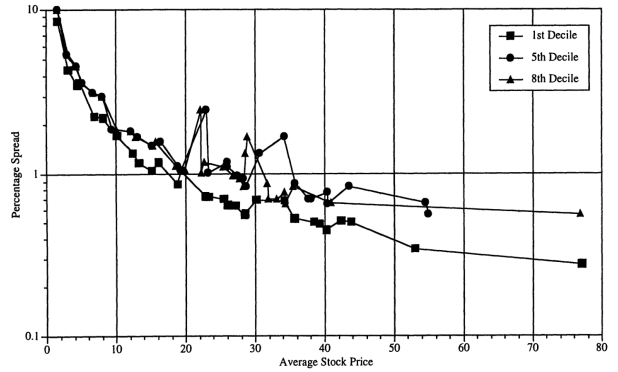


Figure 3: Percentage spread and average stock price by volume decile. This graph shows the relationship between average stock price and average quoted spread as a percentage of the spread midpoint for each of the three 1990 volume deciles in our sample. Average price is calculated as the mean of the CRSP daily file closing price for 10/1/90 to 12/23/90. Average percentage spread is the average of the opening quoted spread divided by the midpoint of the bid/ask quote for the same period as given by the ISSM database.

Figure 3: percentage spread and price

7.4 Table IV: regression results

Read:

- Table IV regresses quoted spread on $V \cdot \text{PIN}$ and dollar volume.
- In the general model, the coefficient on $V \cdot \text{PIN}$ is positive and highly significant.
- The coefficient on dollar volume is negative, consistent with inventory or liquidity effects, but has less explanatory power than $V \cdot \text{PIN}$.
- The adjusted R^2 of the full model is 0.522.
- Restricting the coefficient on $V \cdot \text{PIN}$ to zero reduces the adjusted R^2 to 0.040.
- Restricting the coefficient on volume to zero keeps the adjusted R^2 at 0.413.

Economic message: Information-based trading is much more important than volume alone in explaining quoted spreads.

Table IV
Regression Results

This table presents the results from estimating the linear regression given by $\Sigma = \beta_0 + \beta_1 \cdot V \cdot \text{PI} + \beta_2 \cdot \text{Vol} + \eta$. The dependent variable Σ is the average quoted opening spread calculated from information on the Institute for the Study of Security Markets (ISSM) database for the period of 10/1/90 to 12/23/90. The average price, V , is obtained by averaging the Center for Research in Security Prices (CRSP) daily prices for this same time period. The dollar volume, Vol , is the average daily number of shares traded times V . The probability of informed trade, PI , is reported in the Appendix. The regressions use Ordinary Least Squares (OLS); t -statistics are reported in parenthesis.

	General Model	Restriction to $\beta_1 = 0$	Restriction to $\beta_2 = 0$
Intercept	0.2114 (20.453)	0.2885 (31.954)	0.2034 (18.030)
$V \cdot \text{PI}$	0.0193 (9.463)		0.0178 (7.982)
Vol	-1.035E-11 (-4.572)	-6.879E-12 (-2.175)	
Adj. R^2	0.5216	0.0402	0.4134
F-Value	49.518	4.730	63.720

7.5 Appendix tables: stock-level estimates and sample composition

Read:

- Appendix Table A.I lists the 90 stocks, their matched decile groups, CUSIPs, average prices, and 1990 volumes.
- Appendix Table A.II reports stock-level parameter estimates for the structural model.
- These appendix tables show that the decile-level results are built from stock-by-stock maximum likelihood estimates, not from aggregate decile data.

Economic message: The appendix validates the empirical design: the stocks are price matched but differ sharply in volume, and PIN is estimated separately for each stock.

APPENDIX

Table A.I.
NYSE Stocks Included in Sample

This table presents the stocks included in the testing of our model. Thirty pairs of stocks from the first (highest) 1990 volume decile and the fifth volume decile were selected matched on stock price. An additional sample of thirty stocks from the eighth volume decile with similar prices were added. Average price is calculated as the mean CRSP daily closing price from 10/1/90 to 12/23/90. Total stock volume for 1990 was provided by the NYSE.

Ticker	Company Name	CUSIP	Average Price	1990 Volume
Panel A: First (Highest) Volume Decile Stocks				
CCI	Citicorp	17303410	13.24	409,462,200
WX	Westinghouse Electric Corp.	96040210	27.16	311,279,900
GTE	G T E Corp.	36232010	28.46	298,377,500
F	Ford Motor Co	34537010	28.54	276,229,000
AMD	Advanced Micro Devices Inc.	00790310	4.39	262,823,800
EK	Eastman Kodak Co	27746110	40.20	227,904,900
TOY	Toys R Us	89233510	22.84	204,891,100
UIS	Unisys Corp.	90921410	3.11	199,081,000
PFE	Pfizer Inc.	71708110	77.04	179,367,300
ADM	Archer Daniels Midland Co	03948310	23.25	178,436,550
S	Sears Roebuck & Co	81238710	25.47	163,519,300
AN	Amoco Corp.	03190510	53.00	139,615,500
CHL	Chemical Banking Corp.	16372210	12.38	131,502,700
PN	Pan Am Corp.	69775710	1.52	123,274,200
P	Phillips Petroleum Co	71850710	25.99	121,590,000
DI	Dresser Industries Inc.	26159710	19.34	117,323,200
MXS	Maxus Energy Corp.	57773010	10.09	112,338,700
AL	Alcan Aluminum Ltd.	01371610	18.77	102,497,700
BUD	Anheuser Busch Cos. Inc.	03522910	39.29	102,059,300
PDG	Placer Dome Inc.	72590610	15.14	101,317,310
GT	Goodyear Tire & Rubber Co	38255010	16.03	98,489,100
GPS	Gap Inc.	36476010	34.16	92,811,000
BMG	Battle Mountain Gold Co	07159310	6.84	89,663,700
BR	Burlington Resources Inc.	12201410	42.35	87,124,600
PCI	Paramount Communications Inc.	69921610	38.40	85,805,400
DGN	Data General Corp.	23768810	4.72	84,421,800
CBU	Commodore International Ltd.	20266010	8.06	84,319,400
AET	Aetna Life & Casualty Co	00814010	35.62	81,330,700
BN	Borden Inc.	09959910	30.12	79,998,600
DE	Deere & Co	24419910	43.73	79,562,000
First Volume Decile Mean Values			24.98	154,213,915
Panel B: Fifth Volume Decile Stocks				

Appendix Table A.I, part 1

Table A.I.—Continued

Ticker	Company Name	CUSIP	Average Price	1990 Volume
Panel C—Continued				
LOG	Rayonier Timberlands L P	75507810	19.87	3,400,600
BRY	Berry Petroleum Co	08578910	15.29	3,331,100
ESL	Esterline Technologies Corp.	29742510	6.80	3,317,000
SWN	Southwestern Energy Co	84546710	31.72	3,236,600
WMK	Weis Markets Inc.	94884910	27.79	3,216,000
CRS	Carpenter Technology Corp.	14428510	40.89	3,167,400
PEO	Petroleum & Resources Corp.	71654910	26.81	3,144,000
MSA	Medusa Corp.	58507230	12.78	3,129,300
CMI	Club Med Inc.	18947010	18.54	3,107,300
CRI	Core Industries Inc.	21867510	5.05	3,060,900
CNL	Central Louisiana Elec. Inc.	15389760	34.29	3,056,200
THI	Thomas Industries Inc.	88442510	10.10	3,015,500
CES	Commonwealth Energy Sys	20280010	31.91	2,989,600
Eighth Volume Decile Mean Values			22.82	3,696,783

Table A.II.

Continuous Time Trading Model Parameter Estimates

This table presents the parameters estimated using our model. The parameter μ is the arrival rate of informed traders, ε is the arrival rate of uninformed traders, α is the probability of an information event, and δ is the probability that new information is bad news. The parameter Prob(Inf) is a composite variable measuring the probability of information-based trade. Maximum likelihood estimation is performed using the hill-climbing algorithm GRADX in the GQOPT statistical package. Standard errors are given parenthesis below the parameter estimates.

Ticker	μ	ε	α	δ	Prob(Inf)
Panel A: Estimates for stocks in the first (highest) 1990 volume decile					
ADM	0.148779 (0.006977)	0.253751 (0.002852)	0.477802 (0.067681)	0.191793 (0.077792)	0.122862 (0.015503)
AET	0.101146 (0.006695)	0.164674 (0.002421)	0.414596 (0.074097)	0.373757 (0.103345)	0.112945 (0.017136)
AL	0.055838 (0.003862)	0.066926 (0.001621)	0.536405 (0.078417)	0.503276 (0.096772)	0.182851 (0.021197)
AMD	0.083882 (0.004368)	0.074381 (0.001741)	0.532888 (0.073270)	0.284067 (0.083662)	0.231051 (0.023656)
AN	0.093385 (0.005508)	0.192306 (0.002626)	0.583953 (0.071341)	0.340066 (0.083955)	0.124179 (0.013724)
BMG	0.106477 (0.007582)	0.092259 (0.001623)	0.248948 (0.061586)	0.159629 (0.107645)	0.138763 (0.028324)
BN	0.069492 (0.004492)	0.103059 (0.002494)	0.477719 (0.074444)	0.512118 (0.094444)	0.138719 (0.013719)

Appendix Table A.II, part 1

Table A.I.—Continued

Ticker	Company Name	CUSIP	Average Price	1990 Volume
Panel B—Continued				
SFD	Smiths Food & Drug Ctrs Inc.	83238810	27.53	15,348,300
WOA	Worldcorp Inc.	98190410	4.25	15,157,000
LNC	Lincoln National Corp. In	53418710	38.00	14,820,100
UCU	Utilicorp United Inc.	91800510	19.23	14,701,800
BNL	Beneficial Corp.	08172110	40.41	14,514,100
FMC	F M C Corp.	30249130	28.24	14,338,700
AME	Ametek Inc.	03110510	9.38	14,029,100
HAD	Hadson Corp.	40501810	1.60	14,003,500
FQA	Fuqua Industries Inc.	36102810	12.10	13,884,600
ATM	Anthem Electronics Inc.	03673210	16.43	13,700,500
SCG	Scana Corp.	80589810	34.15	13,649,800
FLO	Flowers Industries Inc.	34349610	13.10	13,528,600
FSS	Firth Sterling Inc.	33799190	28.68	13,417,500
CUM	Cummins Engine Inc.	23102110	35.71	13,400,500
SNG	Southern New England Telecom	84348510	30.68	13,389,600
CKK	Crown Cork & Seal Inc.	22825510	54.94	13,066,800
MCL	Moore Corp. Ltd.	61578510	22.95	13,042,900
MAI	Microwave Association Inc.	55261810	5.11	12,261,900
OGE	Oklahoma Gas & Elec Co	67885810	37.61	12,018,600
LOC	Locite Corp.	54013710	54.51	11,859,000
RGS	Rochester Gas & Elec. Corp.	77136710	18.84	11,704,600
CPY	C P I Corp.	12590210	26.01	11,640,300
EFU	Eastern Gas & Fuel Assoc.	27637710	25.98	11,635,200
GAL	Galoob Lewis Toys Inc.	36409110	2.86	11,620,600
RNB	Republic New York Corp.	76071910	43.48	11,426,800
RLC	Rollins Truck Leasing Corp.	77574110	6.87	10,979,700
Fifth Volume Decile Mean Values			24.17	13,533,110

Panel C: Eighth Volume Decile Stocks

Ticker	Company Name	CUSIP	Average Price	1990 Volume
CZM	Calmat Co	13127110	22.63	4,669,000
HB	Hillenbrand Ind. Inc.	43157310	35.49	4,585,000
LMS	Lamson & Sessions Co	51369610	4.30	4,555,900
SEE	Sealed Air Corp.	82115110	22.21	4,519,700
AJG	Gallagher Arthur J & Co	36357610	22.13	4,395,700
AVA	Audio Video Affiliates Inc.	05090310	3.07	4,381,300
FED	Firstfed Financial Corp.	33790710	15.63	4,324,900
AGL	Angelica Corp.	03466310	28.85	4,319,700
UIC	United Industrial Corp.	91067110	8.01	4,227,200
RXN	Rexene Corp.	76168210	1.59	4,112,400
SAR	Santa Anita Realty Enterprises	80120920	25.47	3,973,800
PNY	Piedmont Natural Gas Inc.	72018610	28.35	3,768,200

Appendix Table A.I, part 2

Table A.II.—Continued

Ticker	μ	ε	α	δ	Prob(Inf)
Panel A—Continued					
CCI	0.224647 (0.007790)	0.416855 (0.003859)	0.627419 (0.064891)	0.358072 (0.079827)	0.144613 (0.013105)
CHL	0.114256 (0.005624)	0.130489 (0.002045)	0.456736 (0.067981)	0.584995 (0.096391)	0.166639 (0.020758)
DE	0.103251 (0.007699)	0.167461 (0.002416)	0.359186 (0.073557)	0.133000 (0.079636)	0.099692 (0.017213)
DGN	0.033160 (0.002614)	0.032406 (0.001248)	0.637148 (0.088598)	0.486356 (0.094280)	0.245841 (0.024250)
DI	0.142132 (0.006148)	0.143106 (0.002082)	0.436768 (0.066102)	0.056801 (0.049985)	0.178239 (0.022343)
EK	0.163630 (0.008890)	0.339637 (0.003320)	0.410295 (0.069020)	0.570004 (0.104349)	0.089946 (0.013605)
F	0.370661 (0.009082)	0.539220 (0.004557)	0.660926 (0.062850)	0.251728 (0.069386)	0.185111 (0.014483)
GPS	0.100973 (0.005220)	0.097646 (0.001773)	0.440241 (0.068166)	0.135349 (0.070289)	0.185416 (0.023345)
GT	0.102726 (0.006685)	0.109990 (0.001806)	0.305960 (0.064026)	0.113204 (0.078136)	0.125014 (0.022522)
GTE	0.157482 (0.006490)	0.287883 (0.003388)	0.658892 (0.067006)	0.060782 (0.040651)	0.152699 (0.013382)
MXS	0.044475 (0.003235)	0.063784 (0.001740)	0.707127 (0.083287)	0.719619 (0.082570)	0.197774 (0.018451)
P	0.064405 (0.004081)	0.122475 (0.002377)	0.727368 (0.074478)	0.571204 (0.081241)	0.160543 (0.014595)
PCI	0.172666 (0.006534)	0.201344 (0.002590)	0.520275 (0.067050)	0.128743 (0.060567)	0.182396 (0.019336)
PDG	0.070785 (0.005077)	0.082074 (0.001768)	0.410499 (0.076262)	0.422886 (0.108068)	0.150395 (0.022452)
PFE	0.150390 (0.007980)	0.252900 (0.003501)	0.504745 (0.079167)	0.373575 (0.091643)	0.130493 (0.016483)
PN	0.092726 (0.004096)	0.074329 (0.001865)	0.647703 (0.071175)	0.790634 (0.066883)	0.287752 (0.021673)
S	0.127108 (0.005659)	0.202135 (0.003185)	0.721529 (0.071523)	0.774888 (0.067537)	0.184910 (0.014379)
TOY	0.210267 (0.010518)	0.257082 (0.003540)	0.450901 (0.074983)	0.037199 (0.036645)	0.155687 (0.020086)
UIS	0.348571 (0.009896)	0.265456 (0.002853)	0.388478 (0.063946)	0.177433 (0.080395)	0.203223 (0.026597)
WX	0.187642 (0.007424)	0.260912 (0.002752)	0.439855 (0.065155)	0.000000 (0.000000)	0.136567 (0.017789)

Panel B: Estimates for stocks in the fifth 1990 volume decile

Ticker	μ	ε	α	δ	Prob(Inf)
AME	0.099409	0.094354	0.506179	0.419519	0.294039

Appendix Table A.II, part 2

8 Short summary

- The paper develops and estimates a structural model of informed and uninformed trade arrivals.
- It measures the probability of information-based trading, PIN, for individual NYSE stocks.
- The sample contains 90 price-matched stocks from the first, fifth, and eighth NYSE volume deciles.
- The most active stocks have lower PIN than less active stocks.
- The fifth and eighth volume deciles have similar PIN.
- Quoted and percentage spreads are lower for the first decile but similar for the fifth and eighth deciles.
- This pattern matches the model's predictions based on information risk.
- Regressions show that $V \cdot \text{PIN}$ has strong explanatory power for spreads, while volume alone has much less power.
- The paper provides an early empirical measure of adverse-selection risk in stock markets.

9 Conclusion

9.1 Core conclusion

Easley, Kiefer, O'Hara, and Paperman show that differences in bid-ask spreads across active and infrequently traded stocks are partly explained by differences in information-based trading risk. The most active stocks have lower PIN and lower spreads. Moderately and less active stocks have similar PIN and similar spreads.

9.2 Economic intuition

The model's intuition is simple. Market makers lose money to informed traders and recover those losses from uninformed traders through the spread. If uninformed trading is abundant, informed trades are diluted and spreads can be low. If uninformed trading is scarce, each trade is more likely to be informed, and spreads must rise.

9.3 Takeaway

The paper turns information risk into an estimable object and shows that this object explains economically important spread differences. Infrequent trading is not only a liquidity problem; it is also an adverse-selection problem.